

39. (g) Implementation Plan: B.Tech. in Mechanical Engineering

Semester - I						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21LEH101T	Communicative English	2	1	0	3	
21MAB101T	Calculus and Linear Algebra	3	1	0	4	
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5	
21MES102L ¹	Engineering Graphics and Design	0	0	4	2	
21EES101T	Electrical and Electronics Engineering	3	1	0	4	
21CYM101T ¹	Environmental Science	1	0	0	0	
21PDM101L ¹	Professional Skills and Practices	0	0	2	0	
21LEM101T ¹	Constitution of India	1	0	0	0	
Total Credits					18	

Semester - II						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21LEH102T	Chinese	2	1	0	3	
21LEH103T	French					
21LEH104T	German					
21LEH105T	Japanese					
21LEH106T	Korean					
21LEH107T	Spanish					
21LEH108T	Russian					
21GNH101J	Philosophy of Engineering	1	0	2	2	
21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4	
21CYB101J	Chemistry	3	1	2	5	
21MES101T	Engineering Mechanics	3	1	0	4	
21CSS101J	Programming for Problem Solving	3	0	2	4	
21BTB103T	Biology	2	0	0	2	
21MES101L ¹	Basic Civil and Mechanical Workshop	0	0	4	2	
21PDM102L ¹	General Aptitude	0	0	2	0	
21GNM101L ¹	Physical and Mental Health using Yoga	0	0	2	0	
21GNM102L ¹	National Service Scheme					
21GNM103L ¹	National Cadet Corps					
21GNM104L ¹	National Sports Organization					
Total Credits					26	

Semester - III						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB201T	Transforms and Boundary Value Problems	3	1	0	4	
21MEC201T	Engineering Thermodynamics	3	0	0	3	
21MEC202T ²	Mechanics of Solids	3	1	0	4	
21MEC203T	Engineering Materials and Metallurgy	3	0	0	3	
21MEC204T	Manufacturing Processes and Metrology	3	0	0	3	
21MEC201L ¹	Manufacturing Processes and Metrology Laboratory	0	0	2	1	
21MEC202L ¹	Material Testing Laboratory	0	0	2	1	
21PDH209T ¹	Social Engineering	2	0	0	2	
21LEM201T ¹	Professional Ethics	1	0	0	0	
21PDM201L ¹	Verbal Reasoning	0	0	2	0	
21LEM202T ¹	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	2	1	0	3	
Total Credits					24	

Semester - IV						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB202T	Numerical Methods	3	1	0	4	
21CSC206T	Artificial Intelligence	2	1	0	3	
21MEC205T ²	Fluid Mechanics and Machinery	3	0	0	3	
21MEC206T	Kinematics and Dynamics of Machines	3	0	0	3	
	E Professional Elective – I				3	
21MEC203L ¹	Machine Dynamics Laboratory	0	0	2	1	
21MEC204L ¹	Fluid Dynamics Laboratory	0	0	2	1	
21MEC205L ¹	Mechanical Modeling and Assembly	0	0	4	2	
21DCS201P ¹	Design Thinking and Methodology	1	2	0	3	
21PDM202L ¹	Critical and Creative Thinking Skills	0	0	2	0	
Total Credits					23	

Semester - V						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB301T	Probability and Statistics	3	1	0	4	
21MEC301T	Thermal Systems Engineering	3	1	0	4	
21MEC301P ¹	Design of Mechanical Systems	3	0	0	3	
21MEC302T ²	Sensors and Control Systems	3	0	0	3	
	Professional Elective – II				3	
O	Open Elective – I				3	
21MEC301L ¹	Thermal Power Systems Laboratory	0	0	2	1	
21MEC302L ¹	Automation and Control Systems Laboratory	0	0	2	1	
21PDM301L ¹	Analytical and Logical Thinking Skills	0	0	2	0	
21LEM301T ¹	Indian Art Form	1	0	0	0	
21GNP301L ¹	Community Connect	0	0	2	1	
Total Credits					23	

Semester - VI						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21CSS303T	Data Science	2	0	0	2	
21MEC301J	Heat and Mass Transfer	3	0	2	4	
21MEC302J ²	Finite Element Methods	3	0	2	4	
21MEC303T	Industry 4.0	3	0	0	3	
	Professional Elective – III				3	
21MEP302L ¹	Project	0	0	6	3	
21MEP303T ¹	MOOC	3	0	0		
	O Open Elective – II				3	
21PDM302L ¹	Employability Skills and Practices	0	0	2	0	
21LEM302T ¹	Indian Traditional Knowledge	1	0	0	0	
Total Credits					22	

Semester - VII						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21GNH401T	Behavioral Psychology	2	1	0	3	
	Professional Elective – IV				3	
	Professional Elective – V				3	
O	Open Elective – III				3	
Total Credits					12	

Semester - VIII						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MEP401L	Major Project	0	0	30	15	
21MEP402L	Major Project	0	0	20	10	
21MEP403L	Internship#	0	0	10	5	
Total Credits					15	

#Students have to register either 21MEP401L or 21MEP402L and 21MEP403L both in eighth semester